

CONGRUENCES FOR THE INTEGRALITY OF $\frac{36n^3 - 19}{m}$

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ABSTRACT. We characterise all pairs $(m, n) \in \mathbb{Z}^2$ for which $m \mid 36n^3 - 19$, equivalently $\frac{36n^3 - 19}{m} \in \mathbb{Z}$. The condition reduces prime-by-prime to the solvability of $36n^3 \equiv 19 \pmod{p}$ for each prime $p \mid m$, which is a cubic-residue problem governed by the splitting of p modulo 3. Using the Chinese Remainder Theorem we construct an explicit, unbounded family of congruence pairs of increasing modulus. The largest pair produced here has a 107-digit modulus.

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1. SETUP AND NECESSARY CONDITIONS

We seek all integers $m \geq 1$ and $n \in \mathbb{Z}$ such that

$$m \mid 36n^3 - 19.$$

By the Chinese Remainder Theorem it suffices to work prime-by-prime: $m \mid 36n^3 - 19$ if and only if $p^{v_p(m)} \mid 36n^3 - 19$ for every prime $p \mid m$, where v_p denotes the p -adic valuation.

Proposition 1.1 (Forbidden primes). *The following primes can never divide $36n^3 - 19$:*

- (1) $p = 2$: $36n^3 - 19 \equiv 1 \pmod{2}$ for all n .
- (2) $p = 3$: $36n^3 - 19 \equiv 2 \pmod{3}$ for all n (in fact $36n^3 - 19 \equiv 8 \pmod{9}$).

Hence any m with $2 \mid m$ or $3 \mid m$ is inadmissible.

Proposition 1.2 ($p = 19$ is special). *For $p = 19$: $19 \mid 36n^3 - 19$ if and only if $19 \mid n$. Moreover, $19^2 \nmid 36n^3 - 19$ for any n . Concretely, if $n = 19k$ then*

$$36(19k)^3 - 19 = 19(36 \cdot 19^2 k^3 - 1),$$

and $36 \cdot 19^2 k^3 \equiv 0 \not\equiv 1 \pmod{19}$, so the 19-adic valuation of $36n^3 - 19$ is exactly 1 whenever $19 \mid n$. The prime 19^2 is therefore permanently forbidden.

2. THE CUBIC RESIDUE CRITERION

For an odd prime $p \notin \{3, 19\}$ with $p \nmid m$, we need $36n^3 \equiv 19 \pmod{p}$, i.e.

$$n^3 \equiv c_p \pmod{p}, \quad c_p := 19 \cdot 36^{-1} \pmod{p}.$$

The number of solutions depends on $p \pmod{3}$.

Theorem 2.1 (Cube roots mod p). *Let $p \geq 5$ be prime, $p \neq 19$, and set $c_p = 19 \cdot 36^{-1} \pmod{p}$.*

- (1) *If $p \equiv 2 \pmod{3}$: the map $x \mapsto x^3$ is a bijection on $\mathbb{F}_p^*$, so there is a unique solution $n \equiv c_p^{(2p-1)/3} \pmod{p}$.*
- (2) *If $p \equiv 1 \pmod{3}$: solutions exist if and only if $c_p^{(p-1)/3} \equiv 1 \pmod{p}$ (the cubic-residue condition). When this holds there are exactly 3 solutions.*

Remark 2.2. For Hensel lifting to prime powers p^k with $k \geq 2$: since the derivative $\frac{d}{dn}(36n^3) = 108n^2 \not\equiv 0 \pmod{p}$ for $p \nmid n$ and $p \nmid 36$, each simple root mod p lifts uniquely to a root mod p^k by Hensel's lemma. The exception is $p = 19$ (covered separately), where no lifting to 19^2 exists.

3. NECESSARY AND SUFFICIENT CONDITIONS

Theorem 3.1 (Complete characterisation). *$m \mid 36n^3 - 19$ if and only if:*

- (1) *m is odd;*
- (2) *$3 \nmid m$;*
- (3) *$19^2 \nmid m$;*
- (4) *for every prime $p \mid m$ with $p \equiv 2 \pmod{3}$, $p \neq 19$: $n \equiv c_p^{(2p-1)/3} \pmod{p}$;*
- (5) *for every prime $p \mid m$ with $p \equiv 1 \pmod{3}$: $c_p^{(p-1)/3} \equiv 1 \pmod{p}$ and n is one of the three cube roots of c_p modulo p ;*
- (6) *if $19 \mid m$: $n \equiv 0 \pmod{19}$.*

The residue class of n modulo m (or $m/19$ if $19 \parallel m$) is then determined by the Chinese Remainder Theorem from conditions (4)–(6).

4. CRT CONSTRUCTION OF CONGRUENCE PAIRS

Fix a finite set $\mathcal{P}$ of admissible primes satisfying the conditions of Theorem 3.1. For each $p \in \mathcal{P}$ choose one residue $r_p \in \{0, \dots, p-1\}$ with $36r_p^3 \equiv 19 \pmod{p}$. By CRT there is a unique $n_0 \pmod{M}$ with $M = \prod_{p \in \mathcal{P}} p$ such that $n_0 \equiv r_p \pmod{p}$ for all p , and $M \mid 36n_0^3 - 19$.

Remark 4.1 (Branching at $p \equiv 1 \pmod{3}$). Each prime $p \equiv 1 \pmod{3}$ where c_p is a cubic residue introduces three admissible residues and hence *triples* the number of CRT branches. All branches are equally valid; we typically fix the lexicographically smallest root for concreteness.

5. EXPLICIT CONGRUENCE PAIRS

5.1. Small cases. The two simplest solutions, which motivated this investigation, are:

- (1) $n \equiv 4 \pmod{5},$
- (2) $n \equiv 10 \pmod{11}.$

Both are verified directly: $36 \cdot 4^3 - 19 = 2285 = 5 \cdot 457$ and $36 \cdot 10^3 - 19 = 35981 = 11 \cdot 3271$.

5.2. The 14-prime congruence. Using the 14 primes $\mathcal{P}_{14} = \{5, 11, 17, 19, 23, 29, 41, 47, 53, 59, 71, 83, 89, 101\}$ with residues $\{4, 10, 1, 0, 19, 15, 22, 23, 18, 15, 15, 76, 29, 97\}$, CRT gives:

- (3) $n \equiv 2094194294170672539119 \pmod{3782234599225650235415}.$

Note that $M_{14} = 3782234599225650235415 = 5 \cdot 11 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 41 \cdot 47 \cdot 53 \cdot 59 \cdot 71 \cdot 83 \cdot 89 \cdot 101$ has 22 digits, and (1), (2) are consequences of (3).

5.3. The 19-prime congruence. Extending $\mathcal{P}_{14}$ by $\{107, 113, 131, 137, 149\}$ (all $\equiv 2 \pmod{3}$), each contributing a unique cube root), with residues $\{13, 19, 50, 122, 142\}$:

$$(4) \quad \begin{aligned} n &\equiv 2061198082866885394580192416899 \\ &\pmod{122289400386770180574091623094795}. \end{aligned}$$

The modulus has 33 digits: $M_{19} = 5 \cdot 11 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 41 \cdot 47 \cdot 53 \cdot 59 \cdot 71 \cdot 83 \cdot 89 \cdot 101 \cdot 107 \cdot 113 \cdot 131 \cdot 137 \cdot 149$.

5.4. Branching at $p = 61$. The prime $61 \equiv 1 \pmod{3}$ is the smallest prime > 3 for which $c_{61} = 19 \cdot 36^{-1} \pmod{61}$ is a cubic residue. Its three cube roots are $\{16, 20, 25\}$. Adjoining $p = 61$ to $\mathcal{P}_{19}$ produces three 20-prime branches, all with modulus $M_{20} = 61 \cdot M_{19} = 7459653423592981015019589008782495$:

$$(5) \quad n \equiv 6850267619741996997543711085725419 \pmod{7459653423592981015019589008782495},$$

$$(6) \quad n \equiv 246639998856407246542763438606489 \pmod{7459653423592981015019589008782495},$$

$$(7) \quad n \equiv 3181585608138891580320962392881569 \pmod{7459653423592981015019589008782495}.$$

5.5. A 25-prime congruence (46-digit modulus). Taking $\mathcal{P}_{25} = \mathcal{P}_{19} \cup \{167, 173, 179, 191, 197\}$ (all $\equiv 2 \pmod{3}$, unique roots $\{75, 127, 124, 105, 93\}$):

$$(8) \quad \begin{aligned} n &\equiv 4019183757032484548523860772875824760867794909 \\ &\pmod{5401692671255252249102543156076416601168074395}. \end{aligned}$$

Modulus M_{25} has 46 digits.

5.6. A 30-prime congruence (58-digit modulus). $\mathcal{P}_{30} = \mathcal{P}_{25} \cup \{227, 233, 239, 251, 257\}$ (unique roots $\{11, 18, 39, 104, 151\}$):

$$(9) \quad \begin{aligned} n &\equiv 1820733127217158956577191662349053768348092988705876831189 \\ &\pmod{5103243448423190660018404944928000789930010683701564743465}. \end{aligned}$$

Modulus M_{30} has 58 digits.

5.7. A 40-prime congruence (83-digit modulus). $\mathcal{P}_{40} = \mathcal{P}_{30} \cup \{263, 269, 281, 293, 311, 317, 347, 353, 359, 383\}$ (roots $\{90, 2, 139, 273, 86, 298, 331, 244, 199, 110\}$):

$$(10) \quad \begin{aligned} n &\equiv \\ &64186161674016772165436788379325330529384310249919785608574714088348016689324291069 \\ &\pmod{} \\ &73002740255141859079375773096078135885056475263749065058307196472256356922583423605. \end{aligned}$$

Both n and M_{40} have 83 digits.

5.8. **A 49-prime congruence (107-digit modulus).** $\mathcal{P}_{49} = \mathcal{P}_{40} \cup \{389, 401, 419, 431, 443, 449, 461, 467, 479, 491\}$
(roots $\{247, 262, 385, 388, 297, 248, 322, 353, 273, 394\}$):

$n \equiv$

$$356677878941720159791604934707498405579918905729307776910365902586590218501760713933634181827 \pmod{M}$$

(11)

$$532426553469205944549230839999202746057882845044670363963998750004013151672896063425700299108$$

Both n and M_{49} have 107 digits.

Remark 5.1. All congruences (1)–(11) have been verified computationally: $(36n^3 - 19) \pmod{M} = 0$ in each case.

6. SUMMARY TABLE

	Label	# primes	Digits of M	Reference
	Small ($p = 5$)	1	1	(1)
	Small ($p = 11$)	1	2	(2)
	14-prime	14	22	(3)
	19-prime	19	33	(4)
	20-prime (3 branches)	20	34	(5)–(7)
	25-prime	25	46	(8)
	30-prime	30	58	(9)
	40-prime	40	83	(10)
	49-prime	49	107	(11)

TABLE 1. Congruence pairs $(n \pmod{M}, M)$ with $M \mid 36n^3 - 19$, ordered by modulus size.

7. STRUCTURAL OBSERVATIONS AND OBSTRUCTIONS

Proposition 7.1 (The moduli grow without bound). *For any $B > 0$ there exists a congruence pair (n_0, M) with $M > B$ and $M \mid 36n_0^3 - 19$.*

Proof. By Dirichlet’s theorem on primes in arithmetic progressions, there are infinitely many primes $p \equiv 2 \pmod{3}$. Each such prime $p \notin \{2, 3\}$ admits a unique r_p with $36r_p^3 \equiv 19 \pmod{p}$. Adjoining further such primes to any existing set $\mathcal{P}$ multiplies the modulus by p while preserving the CRT compatibility, yielding arbitrarily large moduli. $\square$

Proposition 7.2 (Hard obstruction at $p = 2, 3$ and 19^2). *The prime 2, the prime 3, and the prime power 19^2 can never divide $36n^3 - 19$. In particular the 2-part, 3-part, and any 19-part beyond $v_{19} = 1$ of m are permanently forbidden.*

Remark 7.3 (Connection to the Diophantine surface). The variable m (or x in the SageMath search code) here is the divisor parameter in the equation

$$y = -x + \sqrt{(x + 6n)^2 + \frac{36n^3 - 19}{x}},$$

which is equivalent to the factorisation identity

$$x(y - 6n)(y + 2x + 6n) = 36n^3 - 19.$$

This is distinct from the parameter m in the elliptic-surface equation $y^2 = (36n^3 - 19 - 12mn)^2 - (2m)^3$, although the two problems are linked through the same polynomial $36n^3 - 19$.

8. EXTENSION ALGORITHM

The following procedure generates congruence pairs of any desired size.

- (1) Start with $\mathcal{P} = \emptyset$, $M = 1$, $n_0 = 0$.
- (2) Pick any admissible prime $p \notin \mathcal{P}$ (i.e. $p \notin \{2, 3\}$, $p \neq 19^2$).
- (3) Compute $c_p = 19 \cdot 36^{-1} \bmod p$.
 - If $p \equiv 2 \pmod{3}$: compute the unique root $r_p = c_p^{(2p-1)/3} \bmod p$.
 - If $p \equiv 1 \pmod{3}$: check $c_p^{(p-1)/3} \equiv 1 \pmod{p}$; if so, choose any of the three cube roots as r_p . If the condition fails, discard p and return to step 2.
- (4) Set $n_0 \leftarrow \text{CRT}(n_0, r_p; M, p)$ and $M \leftarrow M \cdot p$.
- (5) Verify $M \mid 36n_0^3 - 19$, then repeat from step 2.

Each iteration multiplies M by p , so the modulus grows at least as fast as the product of successive primes $\equiv 2 \pmod{3}$.